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INNOVACIÓN PARA LA EXCELENCIA

OBJECT-ORIENTED PROGRAMMING

TECNIZONE APP

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1. Introduction

This document is a Software Requirements Specification (SRS) for the TecniZoneApp technical service management, inventory control, and sales system. This specification has been structured based on the guidelines provided by the IEEE Recommended Practice for Software Requirements Specifications ANSI/IEEE 830, 1998 standard.

1.1 Purpose

This document defines the functional and non-functional specifications for the development of a web/desktop information system that will manage various operational and commercial processes at TecniZone. It will be used by the General Manager, who is also a workshop technician.

1.2 Scope

This requirements specification is intended for the client and the development team to continue with the system implementation and further automate the store. The system's main objective is to manage administrative and technical processes.

- Registration and receipt of equipment with warranty verification.
- Spare parts inventory control and minimum stock alerts.
- Direct sales, invoicing, and cash control.
- Compilation of monthly financial and performance reports.

1.3 Personnel involved

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1.4 Definitions, acronyms and abbreviations

Name	Description
Tz	TecniZone App
ERS/SRS	Software Requirements Specification (IEEE 830)
RF	Functional Requirement

RNF	Non-Functional Requirement
SystemUser	User with credentials and role within the system
RepairOrder	Technical service order for a specific piece of equipment
StockAlert	Alert generated when inventory falls below minimum

1.5 Summary

This document consists of three sections:

- **First section:** Provides an introduction to the system and an overview of the resource specification.
- **Second section:** Provide a general description of the system (perspective, actors, main functions, constraints and assumptions) without going into excessive code details.
- **Third section:** Define in detail the specific requirements (functional in ERS table format, non-functional and interfaces) that the system must satisfy.

2. Overview

2.1 Product perspective

TecniZoneApp is a centralized information system developed for TECNIZONE, designed to replace the use of static Excel spreadsheets and manually filled-out templates. The system operates using a persistent database that unifies equipment reception, technical service order tracking, spare parts inventory control, sales, and financial performance analysis.

Its main functions restrict and organize access according to staff roles and offer a direct query portal for customers to check the status of their equipment in real time.

2.2 Product functionality

The system includes the following global operational capabilities:

- **PB-01:** Capture and detailed storage of equipment data such as type, brand, model, serial number and faults reported at the time of receipt.
- **PB-02:** Service status management using a standardized drop-down menu.
- **PB-03:** Delivery planning using a visual calendar interface to set estimated delivery dates.
- **PB-04:** Physical verification and control of guarantees using check boxes.
- **PB-05:** History and filtered search of repair orders by customer name or serial number.
- **PB-06:** Critical inventory control of spare parts with automatic text alerts when stock falls below 3 units.
- **PB-07:** Structuring of labor and parts costs with automated calculation of the total on screen.
- **PB-08:** External portal for real-time status inquiry for customers using the order number.
- **PB-09:** Compilation and export of consolidated monthly reports of income and completed repairs.

2.3 User characteristics

User type	Training / Level	Activities and Responsibilities
General Admin	Owner	Monitoring of financial performance
Reception Cashier	Administrative staff	Receiving equipment, initial customer registration, warranty verification, direct sales of spare parts, issuing invoices, and handling inquiries.

Workshop Tech	Specialized support technician	Diagnosis and execution of repairs, updating of order statuses (Under Diagnosis, Under Repair, Listed), manual recording of labor and parts costs, and control of component stock
Client	External client	Initial registration of your devices in store and independent real-time consultation of the status of your equipment by entering your order number.

2.4 Restrictions

- The system is designed under a centralized application architecture connected to a persistent database.
- Stock control requires strict business logic that triggers automatic alerts when the stock level falls below 3 units.
- Order statuses are limited to the three standardized workflow options (Under Diagnosis, Under Repair, Listed).

2.5 Assumptions and dependencies

- It is assumed that the store's computer equipment and terminals have access to the internet or local network to interact with the centralized database simultaneously.
- It is assumed that reception and workshop staff will strictly and truthfully record input data to avoid the information scattering typical of traditional spreadsheets.

3. SPECIFIC REQUIREMENTS

3.1. User Interfaces

The user interface for TecniZoneApp will consist of a structured visual environment designed for clarity, efficiency, and role-based workflows:

- Equipment Reception Form: Features guided input fields and dynamic checkmarks designed for precise physical warranty verification and visual intake logging.
- Service Status Dropdown: A standardized, static dropdown interface for consistent updates across repair stages (Under Diagnosis, Under Repair, Listed).
- Visual Calendar Component: An integrated calendar widget facilitating the accurate selection and commitment of estimated delivery dates.
- Public Status Portal: A streamlined, external web lookup interface allowing end-user customers to track repair progress using a unique order identifier.
- Administrative Dashboard: An operational view incorporating highlighted text notifications for automated low spare parts inventory warnings.

3.1.2 Hardware Interfaces

- Workstation Terminals: Reception and workshop client computers requiring dual-core processors (or superior) and a minimum of 4 GB RAM.
- Printing Equipment: Thermal or standard receipt printers configured for issuing physical job vouchers, invoices, and tracking labels.
- Network Infrastructure: Local Area Network (Ethernet/Wi-Fi) setup facilitating concurrent multi-terminal synchronization with the central database.

3.1.3 Software Interfaces

- Database Management System: Persistent Relational Database Management System (RDBMS) providing unified central storage.
- Web Browsers: Compatibility with modern standard web browsers including Google Chrome, Mozilla Firefox, and Microsoft Edge.

3.4 Communication Interfaces

- Network Protocols: Standard networking protocols (HTTP/HTTPS for external client access and TCP/IP over local network for internal store terminals) ensuring secure data exchange with the persistent database.

3.2 Functional Requirements

Req ID	Requirement Name	Priority	Characteristics	Description
PB-01 / FR01	Device Registration & Unique Identification	High	Identification High Intake technical data capture	The system shall capture and store detailed device information (type, brand, model, serial number, reported faults) in the database via a structured intake form, generating a unique job ID.
PB-02 / FR02	Strategic Service Status Management	High	Normalized static status dropdown	The system shall provide an interface featuring a static dropdown menu (Under Diagnosis, Under Repair, Listed) enabling technicians to manually update the active maintenance phase.
PB-03 / FR03	Delivery Planning & Commitment	Medium	Visual calendar integration	The system shall integrate a visual calendar component into the registration form to select and save the estimated completion date for the technical service.
PB-04 / FR04	Physical Warranty Control & Verification	High	Dynamic condition checkmarks	The system shall implement checkboxes or dynamic checkmarks to verify warranty

				validity and document initial physical intake conditions during equipment reception.
PB-05 / FR05	Historical Tracking & Filtered Search	High	Multi-parameter list filtering	The system shall display a general list of registered jobs with filtering capabilities by customer name or serial number to enable rapid location of past or active orders.
PB-06 / FR06	Critical Inventory & Low-Stock Alerts	High	Automated threshold warnings	The system shall log available spare parts quantities and enforce a business logic rule displaying a warning text message when stock levels drop below 3 units.
PB-07 / FR07	Cost Structuring & Calculation	High	Real-time service charge display	The system shall provide manual entry fields for labor and parts costs, automatically processing the total calculation on screen prior to job closure.
PB-08 / FR08	External Customer Status Portal	High	Authentication-free public lookup	The system shall design a simple search screen where customers can enter their order number to check real-time equipment status without requiring authentication.
PB-09 / FR09	Monthly Management Reporting	Medium	Summary report export	The system shall compile and export a plain text report detailing total revenue collected and completed repair volume within the month (Job Completion &

				ARPR metrics).

3.3 Non-Functional Requirements

3.3.1 Performance Requirements

- NFR01 - Response Time: Cost calculations, status updates, and order filtering queries shall execute within 5 seconds.
- NFR02 - Database Throughput: The persistent database shall support simultaneous queries from reception, workshop technicians, and public portal users without performance degradation.

3.3.2 Security Requirements

- NFR03 - User Authentication & Role Levels: Administrative and workshop capabilities shall require user authentication (SystemUser), strictly enforcing role-based permissions (General Admin, Reception Cashier, Workshop Tech).
- NFR04 - Information Integrity: The public customer portal shall operate strictly in read-only mode, accessible only via order number, to prevent unauthorized data modification or access to internal business records.

3.3.3 Reliability Requirements

- NFR05 - Financial Consistency: Automated cost calculations (including ARPR metrics) shall eliminate manual calculation errors in pricing and invoicing.
- NFR06 - Data Loss Prevention: The persistent RDBMS shall ensure zero loss of repair history, active job logs, or inventory records during system restarts or unexpected closures.

3.3.4 Availability Requirements

- NFR07 - Public Portal Availability: The external customer status portal shall maintain 24/7 continuous availability for customer inquiries.

3.3.5 Maintainability Requirements

- NFR08 - Object-Oriented Architecture: System software shall adhere to a modular object-oriented domain model (SystemUser, Client, Equipment, RepairOrder, Part), ensuring clear documentation, ease of maintenance, and future scalability.

3.3.6 Portability Requirements

- NFR09 - Cross-Platform Accessibility: All web-based system interfaces shall operate seamlessly across major operating systems using modern standard web browsers.